UNIT – I

Simple Mechanisms

Subject : Theory of Machines (23A03403)

Syllabus: Classification of mechanisms, basic kinematic concepts & definitions, degree of freedom & mobility, Grashof's law, inversions of four-bar and slider-crank chains, limit positions, mechanical advantage, transmission angle, quick-return mechanisms, straight-line mechanisms, universal joint, rocker mechanisms.

1. Basic Definitions

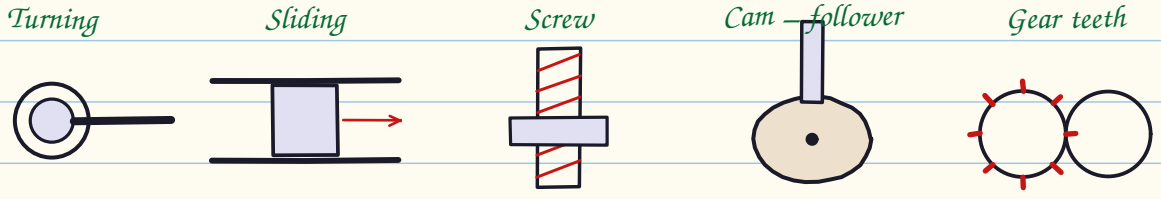
Theory of Machines = study of the relative motion between parts of a machine and the forces acting on them. Divisions : Kinematics (motion without forces), Dynamics (kinetics – forces & motion), Statics.

- **Machine** : combination of resistant bodies that transmits motion and does useful work (engine, lathe). **Mechanism** : a kinematic chain with one link fixed whose purpose is to transmit/transform motion (no concern for energy) – e.g. a clock, typewriter. **Structure** : assembly of resistant bodies with no relative motion.
- **Link (element)** : a resistant body, rigid (assumed) – connecting rod, or flexible (belt, chain) or fluid (hydraulic press). Binary link (2 joints), ternary (3), quaternary (4).
- **Kinematic pair** : two links in contact such that relative motion is constrained.

Classification of kinematic pairs

Basis	Types	Example
Nature of contact	Lower pair (surface contact) / Higher pair (point or line contact)	turning, sliding, screw / cam-follower, gears
Relative motion	Sliding, turning (revolute), rolling, screw (helical), spherical	piston-cylinder, shaft in bearing, ball bearing, nut-bolt, ball & socket

Basis	Types	Example
Closure	Self-closed (geometric) / force-closed (spring, gravity)	journal bearing / cam & follower with spring
Constraint	Completely, incompletely, successfully constrained	shaft in bearing / shaft in foot-step / piston with gas force



lower pair	lower pair	lower pair	higher pair	higher pair
$F = 1$	$F = 1$	$F = 1$	$F = 2$	$F = 2$

- ***Kinematic chain*** : assembly of links and pairs forming a closed loop with relative motion that is constrained. ***Inversion*** : obtaining different mechanisms from the same chain by fixing different links.
- ***Degree of freedom (DOF / mobility)*** : number of independent parameters (inputs) needed to specify the configuration of the mechanism. A rigid body in a plane has 3 DOF ; in space 6.

Classification of mechanisms

- By motion : plane (planar), spherical, spatial. By number of links : 4-bar, 6-bar, 8-bar. By function : function generation, path generation, motion generation. By DOF : single-DOF (constrained), multi-DOF (robot arm).
- By pair type : lower-pair mechanisms (linkages) and higher-pair (cam, gear, ratchet).

2. Degree of Freedom & Mobility

Grübler's (Kutzbach) criterion for plane mechanisms

$$F = 3\,(n - 1) - 2\,j_1 - j_2$$

n = total number of links (incl. frame), j_1 = number of lower pairs (1-DOF), j_2 = number of higher pairs (2-DOF). (For only lower pairs $F = 3(n - 1) - 2j$ — Grübler.)

- $F > 0$: mechanism ; $F = 1$: constrained (single input) ; $F = 0$: statically determinate structure ; $F < 0$: statically indeterminate (superstructure).
- Condition for a chain with only turning pairs having $F = 1$: $j = (3/2)\,n - 2$ and $l = 2p - 4$ (l links, p pairs). Minimum links in a chain = 4 ; the simplest is the four-bar.

Mechanism	n	j	h	F
Four-bar chain	4	4	0	$3(3)-8 = 1$
Single slider-crank	4	4	0	1
Five-bar chain	5	5	0	$3(4)-10 = 2$
Cam + roller follower	3	3	1	$3(2)-6-1 = -1 \rightarrow 1$ (idle roller)
Pantograph / Watt six-bar	6	7	0	$3(5)-14 = 1$

Ex 1. Check the mobility of a mechanism with 8 links and 10 turning pairs : $F = 3(8 - 1) - 2(10) = 21 - 20 = 1$ □ (single DOF). Ex 2. 6 links, 8 pairs : $F = 15 - 16 = -1 \Rightarrow$ structure (locked).

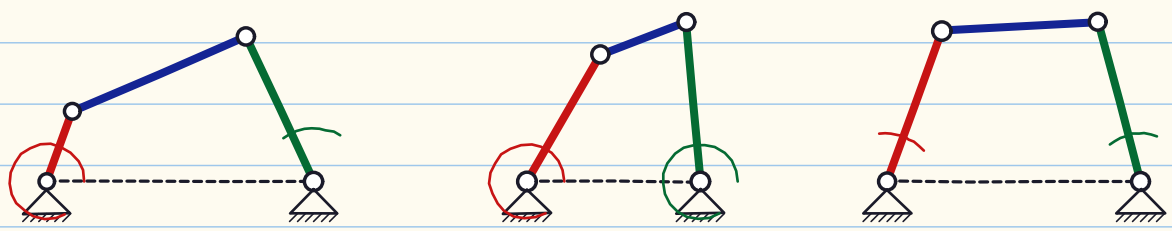
- **Redundant (idle) DOF** : motion that does not affect the output, e.g. rotation of a roller on a cam follower – subtract 1 from the formula. **Over-constrained** mechanisms (parallel links, special dimensions) move even though $F \le 0$ (e.g. parallelogram linkage, locomotive coupling rods).
- **Gruebler's for spatial** (Kutzbach 3-D) : $F = 6(n-1) - 5j_1 - 4j_2 - 3j_3 - 2j_4 - j_5$ (j_i = pairs with i constraints).

3. Four-Bar Chain & Grashof's Law

Crank – rocker
shortest link adjacent to fixed

Double crank
(drag link) – shortest fixed

Double rocker
coupler is shortest



$s + l \le p + q$ (Grashof) — $s + l > p + q$: all double rockers

A four-bar chain has four links (frame d, crank a, coupler b, follower c) connected by four turning pairs.

- Let s = shortest link, l = longest, p and q = other two. **Grashof's law** : for at least one link to make a full revolution relative to the others

$$s + l \leq p + q$$

- If $s + l < p + q$ (Grashof chain), the type depends on which link is fixed :

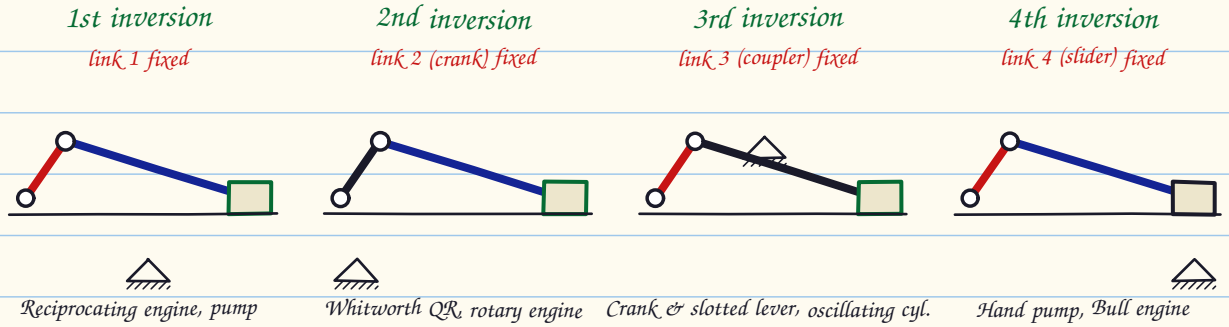
Fixed link	Mechanism	Example
Link adjacent to shortest	Crank-rocker (shortest = crank, other side rocker)	windscreen wiper, sewing machine treadle
Shortest link	Double crank (drag link)	locomotive coupling rods, parallel-ruler
Link opposite to shortest	Double rocker (shortest = coupler)	Watt straight line, Ackermann steering

- If $s + l > p + q$: non-Grashof – every inversion is a double rocker (only oscillation, no complete rotation). If $s + l = p + q$: "change point" – links become collinear, mechanism may flip (parallelogram linkage).

Ex 3. Links 30, 70, 80, 90 mm : $s + l = 30 + 90 = 120$; $p + q = 70 + 80 = 150$ $\Rightarrow 120 < 150 \Rightarrow$ Grashof ; fix 70 (adjacent to 30) → crank-rocker ; fix 30 → drag link ; fix 80 (opposite to 30) → double rocker.

- Special case : **Parallelogram linkage** ($a = c$, $b = d$) – opposite links equal : output rotates at the same speed as input (coupling rods of locomotive; pantograph). **Anti-parallelogram** (crossed) gives speed ratio ≠ 1.

4. Slider-Crank Chain & its Inversions



The single slider-crank chain : 3 turning pairs + 1 sliding pair. Four links : frame (link 1), crank (2), connecting rod (3), slider (4).

1st inversion – link 1 fixed

- Reciprocating engine and compressor : slider (piston) moves in cylinder, converts reciprocation ↔ rotation. Stroke = 2r (crank length r).

2nd inversion – link 2 (crank) fixed

- **Whitworth quick return** : crank fixed, slotted bar rotates fully, ram driven by link – used in shapers and slotters. **Rotary I.C. engine (Gnome)** : cylinders rotate about fixed crankshaft.

3rd inversion – link 3 (connecting rod) fixed

- **Oscillating-cylinder engine, crank & slotted lever quick return** (shaper), and the **slotted-lever** mechanism ; the slider oscillates / rotates about the fixed rod pivot.

4th inversion – link 4 (slider) fixed

- **Hand pump** (link 3 becomes the plunger rod), **Bull engine,** pendulum pump – rod oscillates in the fixed sleeve.

Double slider-crank chain – 2 sliding + 2 turning pairs

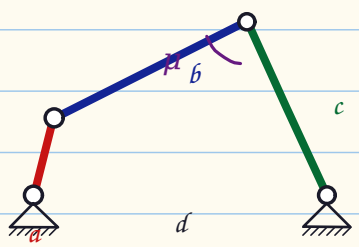
- Elliptic trammel : a point on the connecting bar traces an ellipse (draw ellipses). Scotch yoke : converts rotation to pure SHM of the yoke ($s = r \sin\theta$). Oldham's coupling : connects two parallel shafts with small offset – the intermediate disc slides in grooves, speed ratio 1 : 1.
- Other chains : **Pantograph** (copy to a scale, indicator mechanism), **Geneva (Maltese-cross)** – intermittent motion, **ratchet & pawl, toggle mechanisms.**

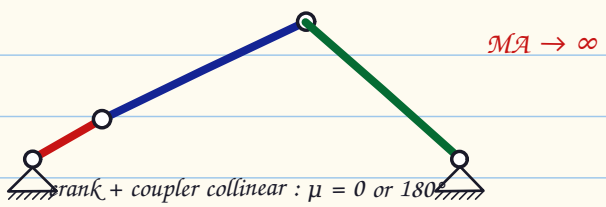
5. Limit Positions, Mechanical Advantage & Transmission Angle

Transmission angle μ

Limit (toggle) positions

$MA = c \sin\mu / (a \sin\delta)$





- **Limit (extreme / dead-centre) positions** : the crank and coupler become collinear (stretched or folded) – the driven link reaches its extreme position and the velocity of the output link is zero. For a slider-crank : inner and outer dead centres ; stroke $H = 2r$. For a crank-rocker the rocker's swing angle ψ is found from these two positions (cosine rule).
- **Transmission angle μ** : acute angle between the coupler and the output (follower) link ; measures the quality of force transmission. Best $\mu = 90°$; poor if $\mu < 40°$ (jamming, high bearing forces).

$$\cos\mu = [\, b^2 + c^2 - a^2 - d^2 + 2ad\cos\theta \,] / (2\,b\,c)\quad \mu_{min} \text{ at } \theta = 0,\ \mu_{max} \text{ at } \theta = 180°$$

- Recommended range $40° \le \mu \le 140°$. Pressure angle (for cams/sliders) $= 90° - \mu$.
- **Mechanical advantage** MA = output torque / input torque $= \omega_{in}/\omega_{out}$ (ideal, no friction). Since the coupler is a two-force member F acts along it : $T_4 = F\,c\sin\mu$, $T_2 = F\,a\sin\delta$ (δ = angle between crank and coupler) :

$$MA = T_4 / T_2 = (c \sin\mu) / (a \sin\delta)$$

- At toggle ($\delta = 0°$ or $180°$) $MA \to \infty$: very large force from small effort (stone crusher, toggle clamp, rock breaker) but motion of the output $\to 0$.

Ex 4. Four-bar $a = 50$, $b = 120$, $c = 80$, $d = 100$ mm. At $\theta = 0$: $\cos\mu = (120^2 + 80^2 - 50^2 - 100^2 + 2(50)(100))/(2\times120\times80) = (14400 + 6400 - 2500 - 10000 + 10000)/19200 = 18300/19200 = 0.953 \Rightarrow \mu = 17.6°$ – poor (μ_{min}) ; at $\theta = 180°$: $\cos\mu = (8300 - 10000)/19200 = -0.0885 \Rightarrow \mu = 95.1°$ (μ_{max}).

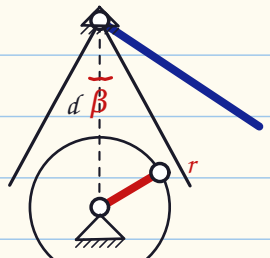
Stroke & time ratio of a slider-crank

- Time ratio (offset slider) $Q = (180° + \beta)/(180° - \beta)$, β = angle between the two lines of coupler at dead centres ($\beta = 0$ if no offset $\Rightarrow$ equal times). Stroke $H = 2r$ for in-line.

6. Quick-Return Mechanisms

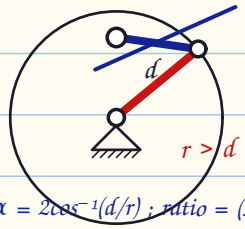
Used in shapers, slotters, presses – the cutting stroke should be slow and uniform, return stroke fast to save time.

Crank & slotted lever



$\sin(\beta/2) = r/d$; ratio = $(180°+\beta)/(180°-\beta)$

Whitworth quick return



$\alpha = 2\cos^{-1}(d/r)$; ratio = $(360°-\alpha)/\alpha$

cutting stroke : slow
return stroke : fast

Crank & slotted lever (3rd inversion)

- Fixed distance d between the crank pivot and the lever pivot, $d > r$. The lever swings between two tangent positions through angle β :

$$\sin(\beta/2) = r/d \quad \text{Time ratio} = \text{cutting time / return time} = (180° + \beta)/(180° - \beta)$$

- Length of stroke = 2 × (lever length to ram link) × $\sin(\beta/2)$ = L (ram arm) × (chord) ; ram stroke = $2R\sin(\beta/2)$ where R = length of the swinging arm (pivot to ram pin).

Ex 5. r = 100 mm, d = 250 mm : $\sin(\beta/2) = 0.4 \Rightarrow \beta = 47.2°$; time ratio = 227.2/132.8 = 1.71.

Whitworth quick return (2nd inversion)

- Crank fixed (O) is the fixed link of length d, driving crank length $r > d$. Lever pivot P at distance d from O ; slotted lever makes full revolution.

$$\alpha = 2\cos^{-1}(d/r) \quad \text{Time ratio} = (360° - \alpha)/\alpha = \text{cutting angle / return angle}$$

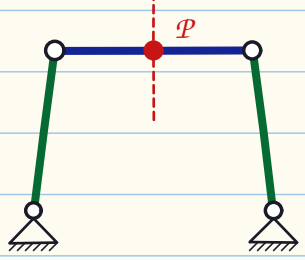
Ex 6. r = 150 mm, d = 100 mm : $\alpha/2 = \cos^{-1}(0.667) = 48.2° \Rightarrow \alpha = 96.4°$; ratio = (360 − 96.4)/96.4 = 2.73.

- Other QR : **Drag-link mechanism** (double-crank, shortest link fixed) – ratio up to 3 ; offset slider-crank.

7. Straight-Line Mechanisms

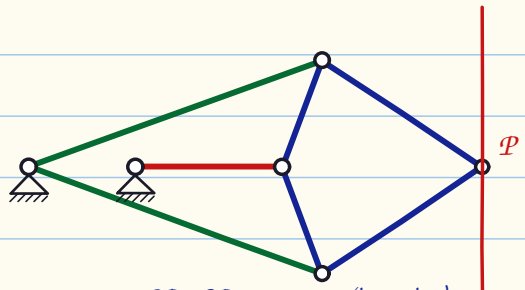
Convert rotary motion to straight-line motion using only lower pairs (no guides) – exact or approximate.

Watt (approximate)



P near mid-point of coupler

Peaucellier (exact straight line)

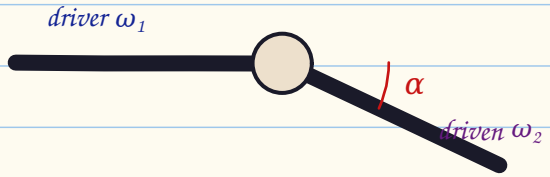


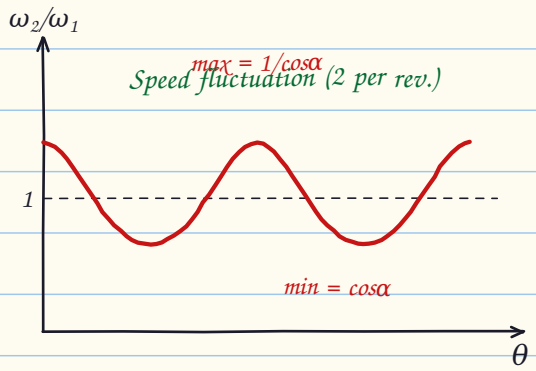
OB · OP = constant (inversion)

- **Exact** : Peaucellier (8 links – an inversor : OB · OP = const. so B moves on a circle through O, P on a straight line) ; Hart's (6 links, anti-parallelogram) ; Scott-Russell (a slider moves a bar with a mid-point fixed pivot – the end moves in a line perpendicular).
- **Approximate** : Watt (double rocker – P at the coupler mid-point moves on a figure-8 with a nearly straight part, used in steam engine indicator), Grasshopper, Tchebicheff, Robert's.
- Condition for the Peaucellier cell : $AB = BC = CD = DA$ (rhombus), $OA = OC$, and $OO_1 = O_1B$. Then $OB \times OD = OA^2 - AB^2$ = constant.
- Straight-line motion is needed in engine indicators, machine tools, sheet-metal handling and robotic arms.

8. Universal (Hooke's) Joint

Hooke's (universal) joint





A joint connecting two intersecting shafts (inclined at α) – used in automobile propeller shafts, rolling mills, machine tools. Two forks at right angles joined by a cross (spider) – a spherical four-bar linkage.

- If the driving shaft turns through θ and the driven through φ :

$$\tan\theta = \cos\alpha \cdot \tan\varphi \quad \omega_2/\omega_1 = \cos\alpha \,/\, (1 - \cos^2\theta \sin^2\alpha)$$

- Speed ratio varies twice per revolution : $\omega_{2,max} = \omega_1/\cos\alpha$ ($\theta = 0°, 180°$) ; $\omega_{2,min} = \omega_1\cos\alpha$ ($\theta = 90°, 270°$).

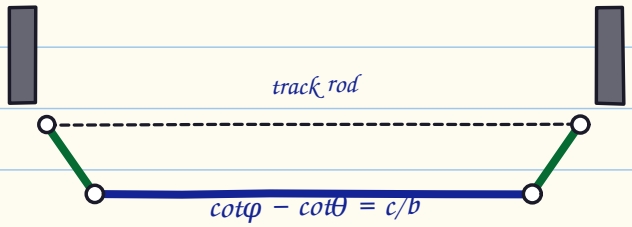
Coefficient of fluctuation of speed = $(\omega_{2max} - \omega_{2min})/\omega_1 = \tan\alpha \sin\alpha$; mean speed equal to ω_1.

- A larger α gives larger speed fluctuation, inertia torques and vibration ; practical $\alpha \le 15°–20°$.
- **Double Hooke's joint** : two joints with intermediate shaft ; if the driving & driven shafts make equal angles with the intermediate shaft and the forks of the intermediate shaft are in the same plane, $\omega_3 = \omega_1$ – constant-velocity transmission (automobile drive shaft, steering columns). Constant-velocity joints (Rzeppa, Tracta) also used.

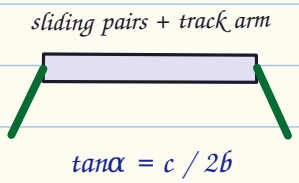
Ex 7. $\alpha = 20°$: $\omega_{2max} = \omega_1/0.9397 = 1.064\ \omega_1$; $\omega_{2min} = 0.9397\ \omega_1$; fluctuation = $\tan20° \sin20° = 12.4\ \%$. For $\omega_1 = 600$ rpm : 563.8 – 638.5 rpm.

9. Rocker Mechanisms & Steering Gears

Ackermann steering gear



Davis steering gear



- **Crank-rocker** – input crank rotates, output rocker oscillates : wipers, rocking-chair, radar scanners, looms. **Rocker-rocker** (double-rocker) – both oscillate : Ackermann steering, aircraft landing-gear. **Rocker-slider** – oscillating arm drives a slider : valve gear, shaper ram drive.
- **Steering gear condition** : for pure rolling of all wheels in a turn, axes of all wheels must meet at one point (instantaneous centre) – the inner wheel turns more than the outer :

$\cot\varphi - \cot\theta = c / b$ (φ outer, θ inner, c = distance between pivots, b = wheel base)

- **Davis steering gear** has sliding pairs and satisfies the condition exactly ($\tan\alpha = c/2b$) but wears quickly. **Ackermann** – only turning pairs, exact at straight-ahead and at one turning angle, approximately in between – the common choice.

Ex 8. Ackermann gear : wheel base $b = 2.4$ m, pivot distance $c = 1.2$ m. For inner wheel $\theta = 30°$: $\cot\varphi = \cot 30° + c/b = 1.732 + 0.5 = 2.232 \Rightarrow \varphi = 24.1°$ (outer wheel).

10. Other Common Mechanisms (short notes)

- **Pantograph** : four links forming a parallelogram with extended arms ; a point traces a path similar to another point to a fixed scale (ratio = OB/OA). Uses : copying drawings, engraving, indicator rigs, current collector on electric trains.
- **Geneva (Maltese-cross) drive** : a driving pin enters a slot of the driven wheel once per revolution and indexes it by $360°/n$; the wheel is locked in between by the circular disc – used in film projectors, indexing tables. For n slots the driver turns through $180° - 360°/n$ during indexing.
- **Ratchet & pawl** : intermittent rotation in one direction only (jacks, winches, ratchet spanner, clock escapement). **Toggle mechanism** : as the links become collinear a small effort gives very large force (stone crusher, riveting, clamps, punching).
- **Scotch yoke** displacement $x = r\sin\theta$, velocity $r\omega\cos\theta$, acceleration $-r\omega^2\sin\theta$ – pure SHM. **Rapson's slide** and **Corliss gear** are used in steam-ship steering and valve control.
- **Equivalent linkages** : a higher pair (cam-follower / gear contact) can be replaced for kinematic analysis by an equivalent lower-pair four-bar whose links join the centres of curvature of the mating profiles.
- **Kinematic diagram** : a skeleton of links drawn to scale showing only the dimensions that affect the motion – first step of every analysis (displacement, velocity, acceleration).

11. Additional Solved Problems

P1 (mobility). Plane mechanism with 7 links, 8 lower pairs and 1 higher pair : $F = 3(7-1) - 2(8) - 1 = 18 - 16 - 1 = 1$.

P2 (Grashof, change point). Links 40, 60, 100, 100 : $s + l = 140$, $p + q = 160 \Rightarrow s + l < p + q \Rightarrow$ Grashof □. Links 40, 60, 100, 120 mm : $s + l = 160 = p + q \Rightarrow$ change-point mechanism.

P3 (shaper design). Stroke of ram 250 mm, cutting : return ratio 1.8, crank & slotted lever. $(180 + \beta)/(180 - \beta) = 1.8 \Rightarrow 2.8\beta = 144 \Rightarrow \beta = 51.4°$; $\sin(\beta/2) = 0.434 \Rightarrow r/d = 0.434 \Rightarrow$ for $d = 300$ mm : $r =$ 130 mm ; swing arm $R = 250/(2 \times 0.434) =$ 288 mm.

P4 (slider-crank). Crank 120 mm, connecting rod 480 mm, in-line : stroke $= 2r = 240$ mm ; $n = l/r = 4$. Dead centres at distance $l + r = 600$ mm and $l - r = 360$ mm from the crank pivot (difference = stroke □).

P5 (transmission angle). A four-bar with $a = 20$, $b = 70$, $c = 60$, $d = 80$ mm : at $\theta = 0$, $\cos\mu = (4900 + 3600 - 400 - 6400 + 3200)/(2\times70\times60) = 4900/8400 = 0.583 \Rightarrow \mu_{min} =$ 54.3° ; at 180° : $\cos\mu = (1700 - 3200)/8400 = -0.179 \Rightarrow \mu_{max} = 100.3°$ (within 40°–140° □).

Short answers – Quick Revision

- **Difference between machine and mechanism?** A mechanism only transmits/modifies motion ; a machine also transforms energy and does work.
- **Why can a structure not be a mechanism?** $F \le 0$ – no relative motion possible between the links.
- **Why are higher pairs necessary?** Cams and gears give required non-uniform motion / speed ratios that lower pairs cannot, at the cost of high contact stress.
- **Why quick return needed?** Idle return stroke reduces non-productive time ; the cutting stroke takes the larger crank angle (speed uniform).
- **What is an inversion?** Fixing a different link of the same chain ⇒ different mechanism with the same relative motion between links.
- **Which inversion gives a slow cutting stroke and fast return?** Crank and slotted lever / Whitworth – 3rd and 2nd inversions of the single slider-crank.
- **Why is the double Hooke's joint used?** Cancels the speed fluctuation of a single joint ($\omega_1 = \omega_3$ when forks of intermediate shaft are in line).

12. Quick Formula Sheet – Unit I

Item	Formula
Mobility	$F = 3(n-1) - 2j_1 - j_2$
Grashof	$s + l \le p + q$
Transmission angle	$\cos\mu = (b^2+c^2-a^2-d^2+2ad\cos\theta)/2bc$

Item	Formula
Mechanical advantage	$MA = c \sin\mu / (a \sin\delta)$
Crank-slotted QR	$\sin(\beta/2) = r/d$; ratio $= (180+\beta)/(180-\beta)$
Whitworth QR	$\alpha = 2\cos^{-1}(d/r)$; ratio $= (360-\alpha)/\alpha$
Hooke's joint	$\tan\theta = \cos\alpha \tan\varphi$; $\omega_2/\omega_1 = \cos\alpha/(1-\cos^2\theta \sin^2\alpha)$
Steering	$\cot\varphi - \cot\theta = c/b$

13. Important Questions

- Define kinematic pair, link, kinematic chain, mechanism and inversion. Classify kinematic pairs with examples.
- Derive Grübler's criterion and find the DOF of the given linkages.
- State Grashof's law. Explain the inversions of the four-bar chain with sketches and applications.
- Explain the inversions of a single slider-crank chain and the double slider-crank chain.
- What is transmission angle? Derive the expression for mechanical advantage of a four-bar.
- Explain the working of Whitworth quick return and crank & slotted lever mechanisms; derive the time ratio.
- Explain the Peaucellier and Watt mechanisms. Derive the speed ratio of a Hooke's joint; double Hooke's joint.
- Explain Ackermann & Davis steering gears with the condition for correct steering.